

125A Plugin Tester - User Guide

Version 0.2.0 - Windows x64 - Verified Build #61

Overview

125A Plugin Tester is a technical QA and release-checking tool for VST3 plug-ins on Windows. It combines custom structure, state, processing, I/O and stress tests with the official Steinberg VST3 Validator and produces traceable TXT and JSON reports.

Usage

1. Start 125A_Plugin_Tester.exe.
2. Select the VST3 plug-in to test.
3. Let the complete test run finish.
4. Review the main report and, if required, the detailed probe reports. PASS, WARNING, FAIL and the final release recommendation are the key results.

Test areas

The tester checks module/factory/component/controller behavior, state roundtrip and fresh-instance behavior, processing configurations across multiple sample rates and buffer sizes, silence/impulse/DC/denormal scenarios, sustained processing and lifecycle behavior, plus isolated I/O, sidechain and event contracts. The Steinberg VST3 Validator is also executed.

Reports and interpretation

PASS: test passed.

WARNING: an issue worth reviewing, but not necessarily release-blocking.

FAIL: reproducible technical failure; release is not recommended.

TXT reports are intended for direct technical review; JSON reports are intended for machine processing.

Verified reference test

Version 0.2.0 was verified with an intentionally faulty BadIO VST3 fixture. The 125A I/O probe correctly detected: **Bus arrangement - out-of-range bus index was accepted**. The main report surfaced that exact failure. The same fixture simultaneously passed all 47 Steinberg Validator tests, practically verifying the additional 125A check for this deliberately injected contract violation.

Windows warning: unknown publisher / SmartScreen

The current 125A_Plugin_Tester.exe is not signed with a publicly trusted Windows code-signing certificate. Windows may therefore show **Windows protected your PC, Unknown publisher**, or a similar SmartScreen warning on first launch. This is expected for a new or unsigned desktop executable. The warning by itself does not state that the file contains malware.

How to run the current unsigned build

Only proceed when the EXE came from the official 125A repository.

SmartScreen: Select **More info** and then **Run anyway** if Windows offers that option.

Unblock the file: Right-click 125A_Plugin_Tester.exe -> **Properties** -> if present, enable **Unblock** in the Security section -> Apply. For a file you have intentionally verified, PowerShell can also be used: **Unblock-File -Path .\125A_Plugin_Tester.exe**.

On managed corporate PCs or systems with Windows Smart App Control enabled, unsigned apps can be blocked by

policy. Do not bypass an organization's security policy.

Permanent fix for future releases

The correct long-term fix is an **Authenticode code signature** backed by a publicly trusted publisher identity. Options include a valid OV/EV code-signing certificate or Microsoft Artifact Signing (formerly Trusted Signing), where available to the developer. The final EXE should be signed with SHA-256 and RFC 3161 timestamped after the release build. Every release should use the same verified publisher identity consistently. A valid signature does not guarantee that a brand-new file will never show SmartScreen immediately, because Microsoft also builds file-hash and publisher reputation. It does provide a verified publisher and is the proper permanent solution.

Example release process

1. Set up a code-signing certificate or Artifact Signing profile.
2. Build the release EXE.
3. Sign the EXE with SHA-256 using SignTool/Artifact Signing.
4. Add an RFC 3161 timestamp.
5. Verify the signature before upload, for example with Windows Properties or SignTool verify.
6. Publish that exact signed binary and keep using the same signing identity for subsequent releases.

System and license

Target platform: Windows x64; plug-in format: VST3. This project uses the Steinberg VST3 SDK under its applicable licensing terms. 125A project source code: all rights reserved unless explicitly stated otherwise.

Note: Microsoft guidance in 2026 states that even correctly signed new apps may need time to build SmartScreen reputation; EV signatures no longer automatically bypass SmartScreen.

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